

V4.2 Toy Model 1D-C1

Scalar Delay + Vectorial Connectivity

Purpose:

Test Catalin's vectorial correction: scalar rules are too blunt; direction matters.

Cost rule:

$$c_{ij} = c0_{ij} * [1 + \lambda R_{\lambda}] / [1 + \mu R_{\mu} (\hat{e} \cdot \hat{g})^2]$$

Formula meaning:

λ = scalar delay sensitivity; it increases cost in the wide delay field.

μ = vectorial connectivity sensitivity; it decreases cost only when the edge direction aligns with the radial gradient direction. The squared dot product preserves symmetry.

Field radii:

$\rho_{\lambda} = 4.0$: wide delay field.

$\rho_{\mu} = 2.0$: tight connectivity field.

Diagnostics:

L = scout delay relative to flat baseline.

$F_{center} > 0$ = off-axis paths moved inward near the centre.

$Detour > 0$ = off-axis paths moved outward near the centre.

Inner paths = number of bundle paths entering the tight connectivity radius.

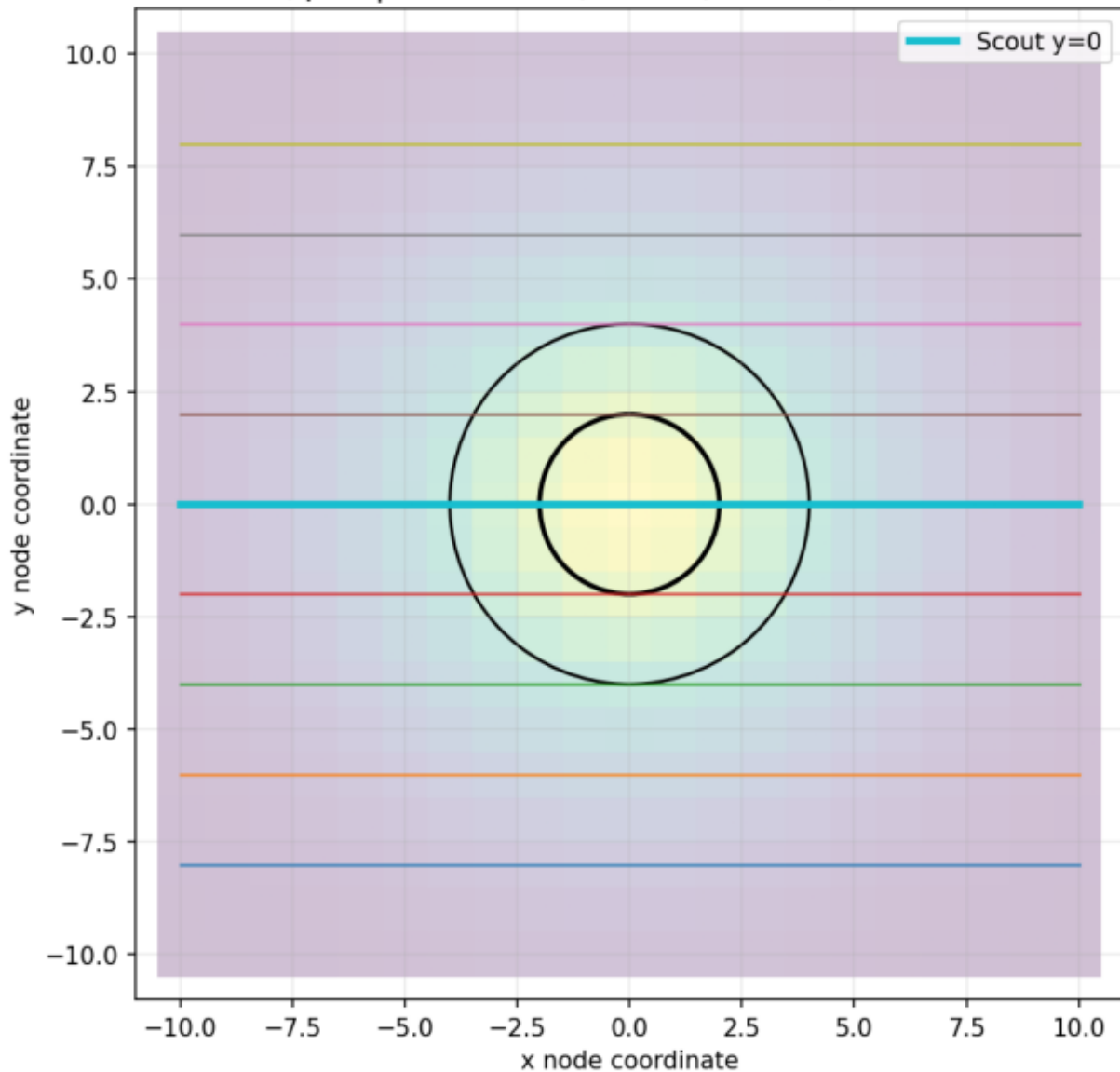
Guardrail:

This is not General Relativity and not gravitational lensing. It is a graph-based test of whether directional relational connectivity can produce controlled focusing-like behaviour.

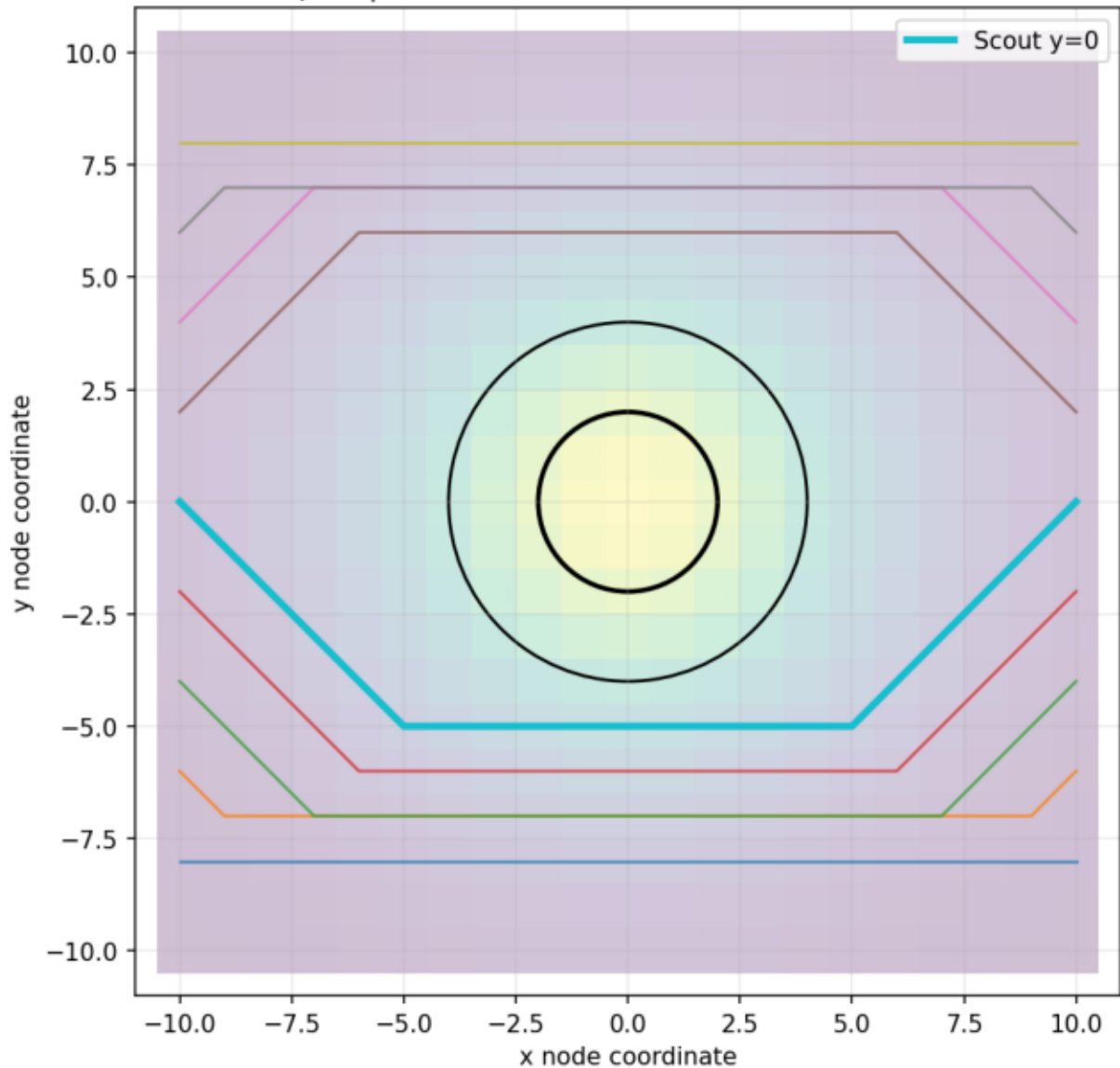
Numerical summary:

C1_flat_control: $\lambda=0, \mu=0, scout=20.000, L=0.000, avgDelay=0.000, F=0.000, detour=0.000, inner=3/9$
C1_delay_only_l1: $\lambda=1, \mu=0, scout=29.711, L=9.711, avgDelay=4.674, F=-2.000, detour=2.000, inner=0/9$
C1_connect_only_m3: $\lambda=0, \mu=3, scout=14.631, L=-5.369, avgDelay=-2.040, F=3.000, detour=-3.000, inner=7/9$
C1_l1_m1: $\lambda=1, \mu=1, scout=24.235, L=4.235, avgDelay=3.839, F=-0.500, detour=0.500, inner=3/9$
C1_l1_m2: $\lambda=1, \mu=2, scout=21.622, L=1.622, avgDelay=2.968, F=-0.500, detour=0.500, inner=3/9$
C1_l1_m3: $\lambda=1, \mu=3, scout=20.013, L=0.013, avgDelay=2.182, F=1.250, detour=-1.250, inner=5/9$
C1_l2_m3: $\lambda=2, \mu=3, scout=25.395, L=5.395, avgDelay=5.161, F=-1.750, detour=1.750, inner=3/9$
C1_l2_m5: $\lambda=2, \mu=5, scout=22.567, L=2.567, avgDelay=4.218, F=-1.750, detour=1.750, inner=3/9$

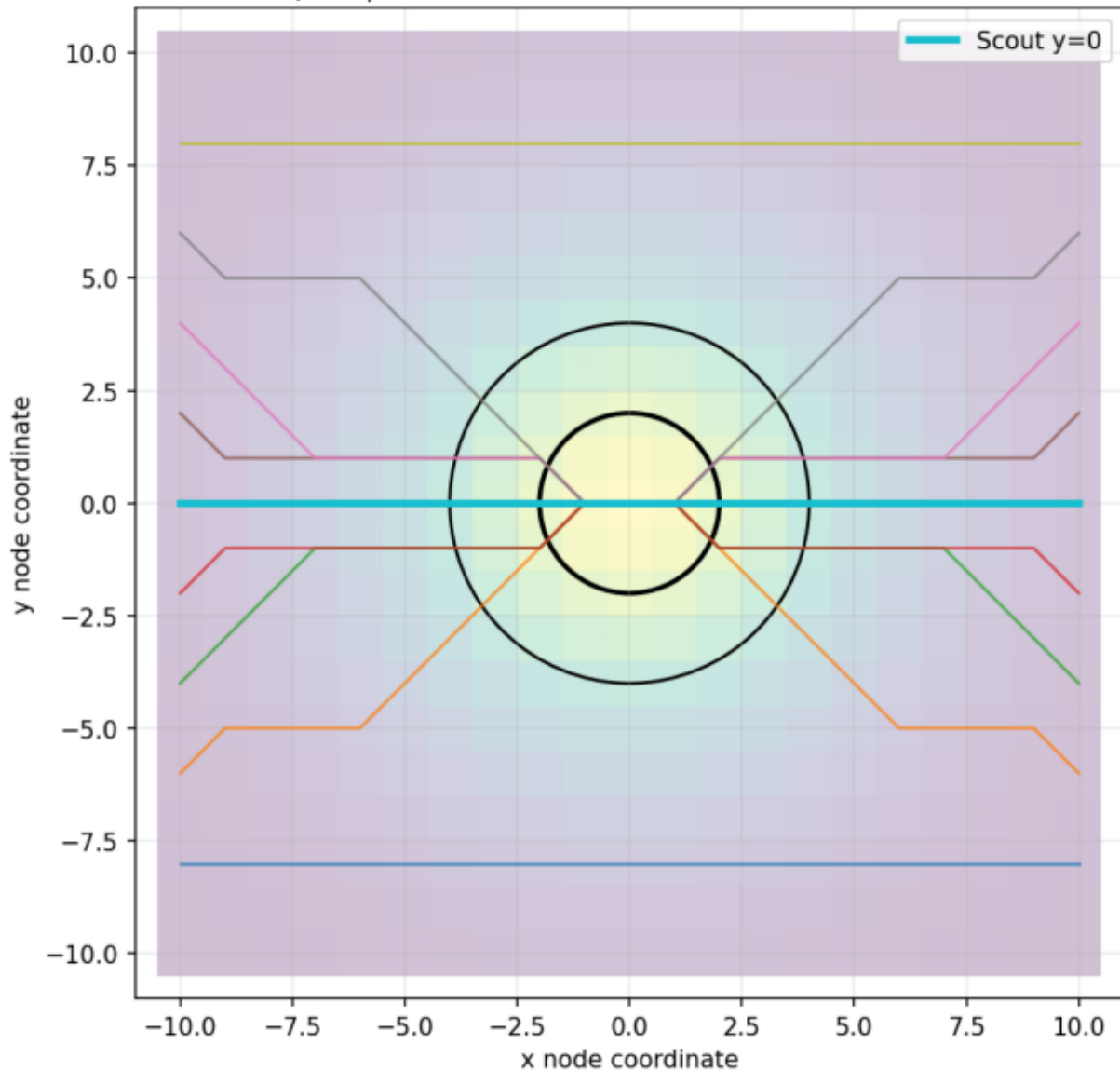
C1_flat_control
 $\lambda=0, \mu=0$ | scout=20.00, L=0.00, F=0.00, Detour=0.00



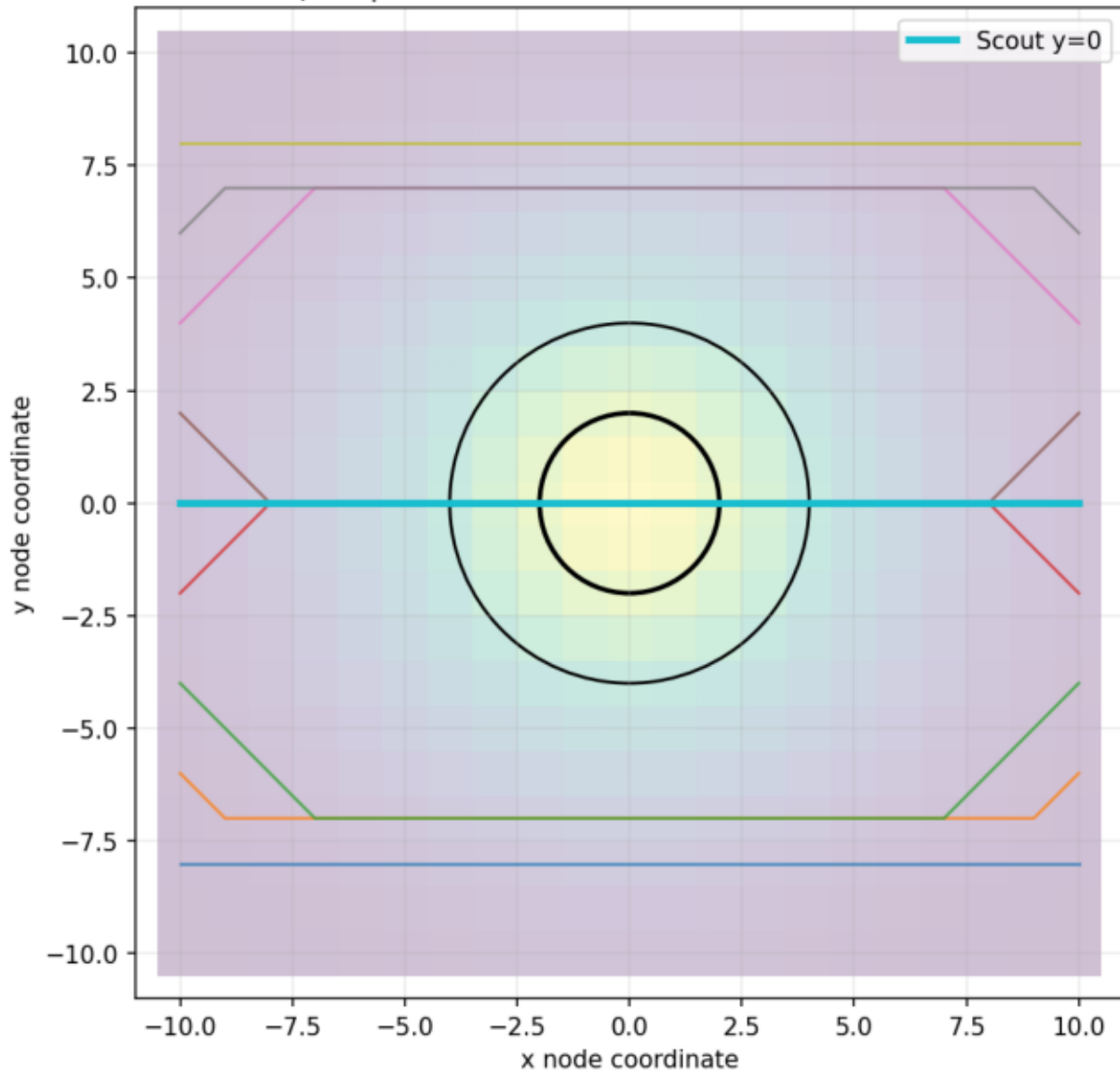
C1_delay_only_l1
 $\lambda=1, \mu=0$ | scout=29.71, $L=9.71$, $F=-2.00$, Detour=2.00



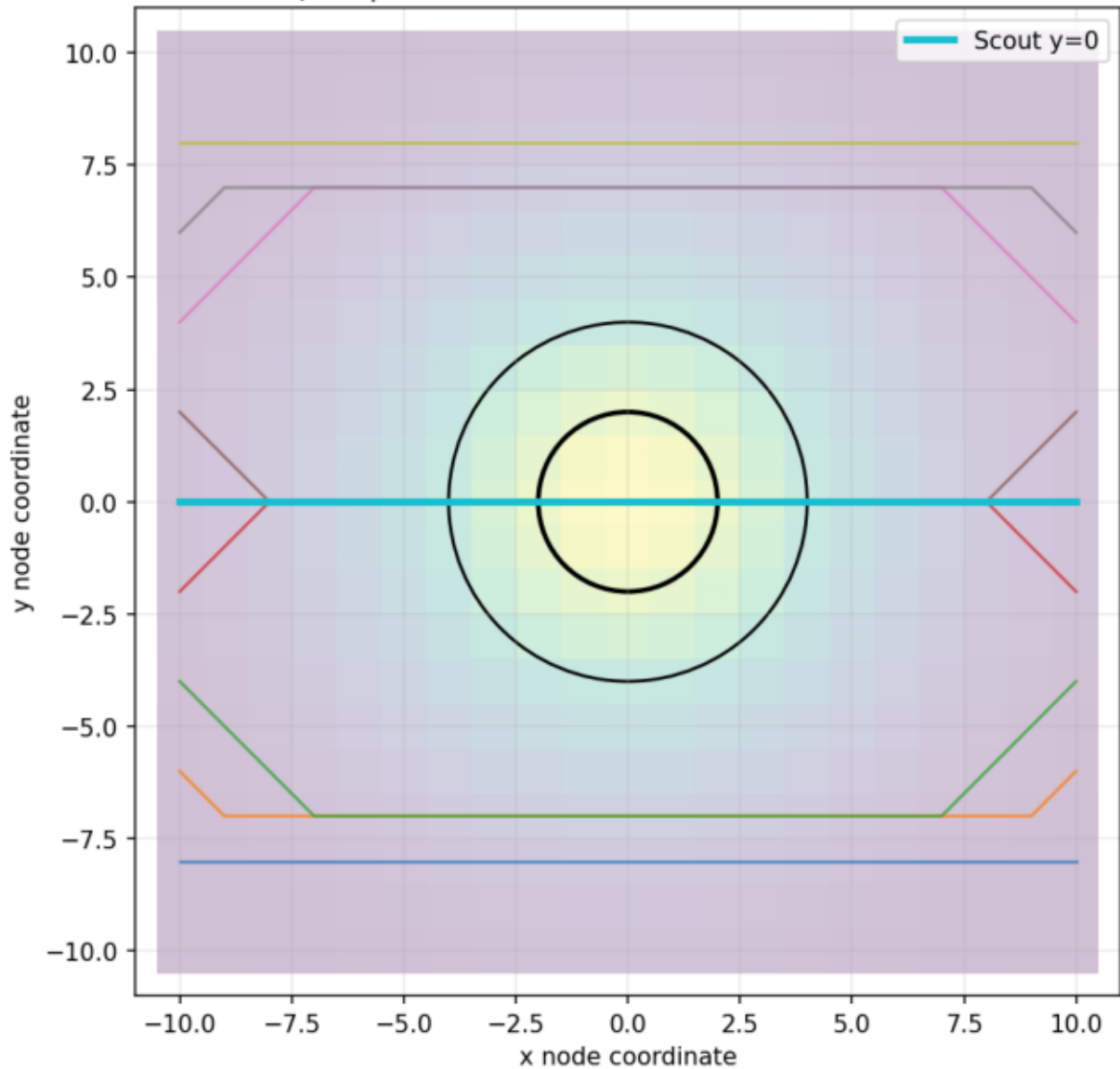
C1_connect_only_m3
 $\lambda=0, \mu=3$ | scout=14.63, $L=-5.37$, $\bar{F}=3.00$, Detour=-3.00



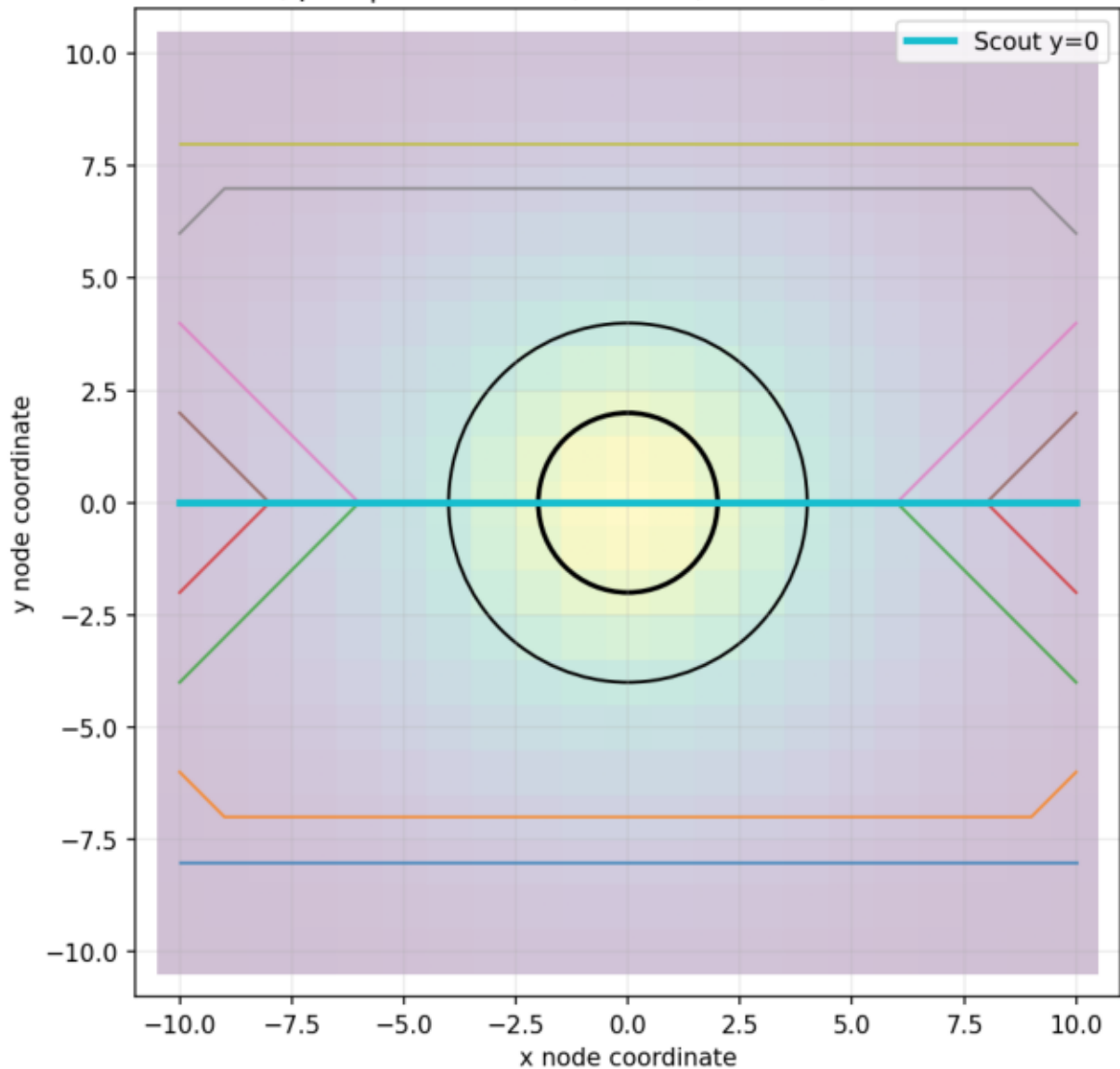
C1_l1_m1
 $\lambda=1, \mu=1$ | scout=24.24, $\bar{L}=4.24$, $F=-0.50$, Detour=0.50



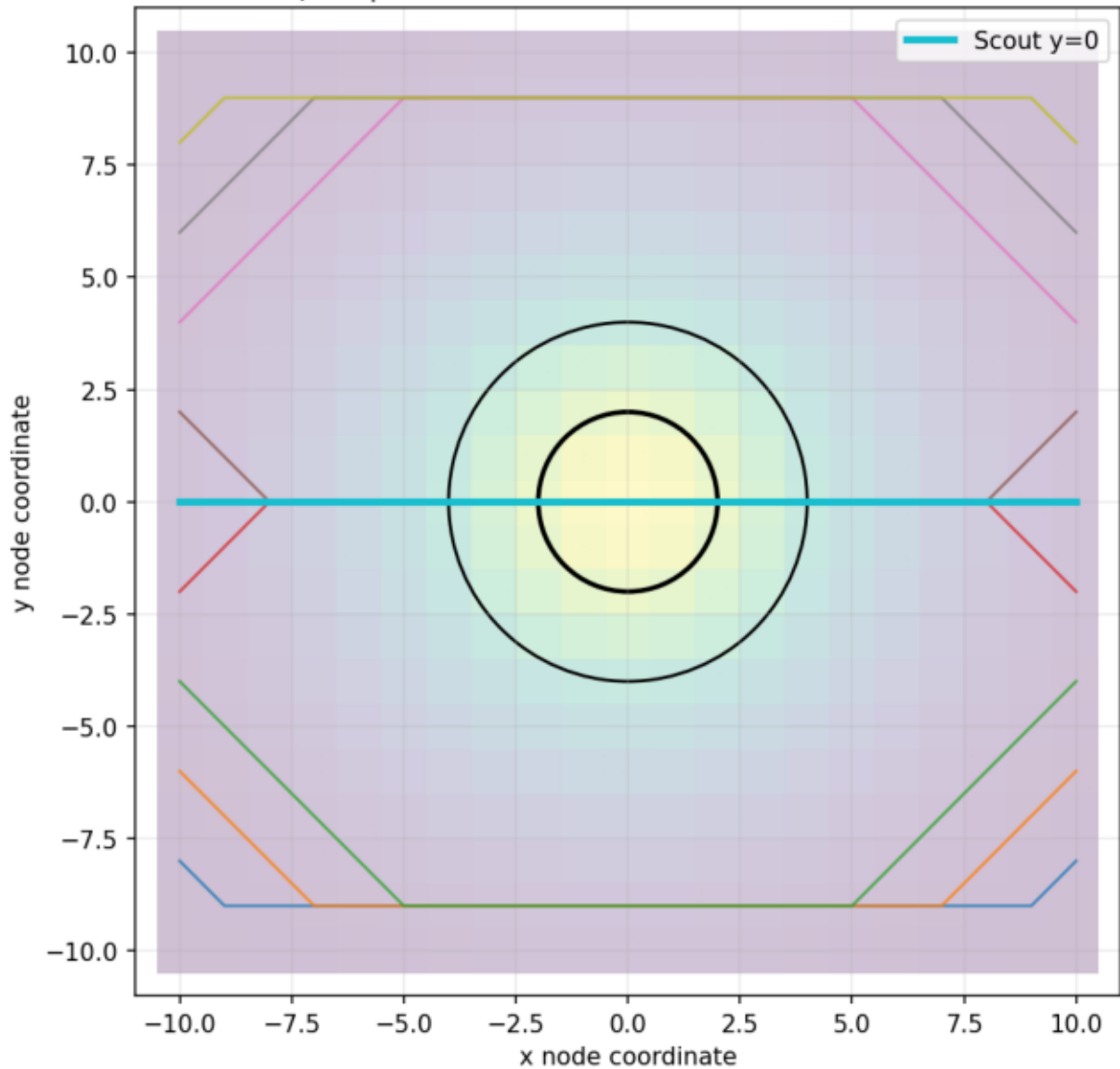
$C1_l1_m2$
 $\lambda=1, \mu=2 \mid \text{scout}=21.62, \bar{L}=1.62, F=-0.50, \text{Detour}=0.50$



C1_l1_m3
 $\lambda=1, \mu=3$ | scout=20.01, $\bar{L}=0.01$, $F=1.25$, Detour=-1.25



C1_l2_m3
 $\lambda=2, \mu=3$ | scout=25.40, $\bar{L}=5.40$, $F=-1.75$, Detour=1.75



C1_l2_m5

$\lambda=2, \mu=5$ | scout=22.57, $\bar{L}=2.57$, $F=-1.75$, Detour=1.75

